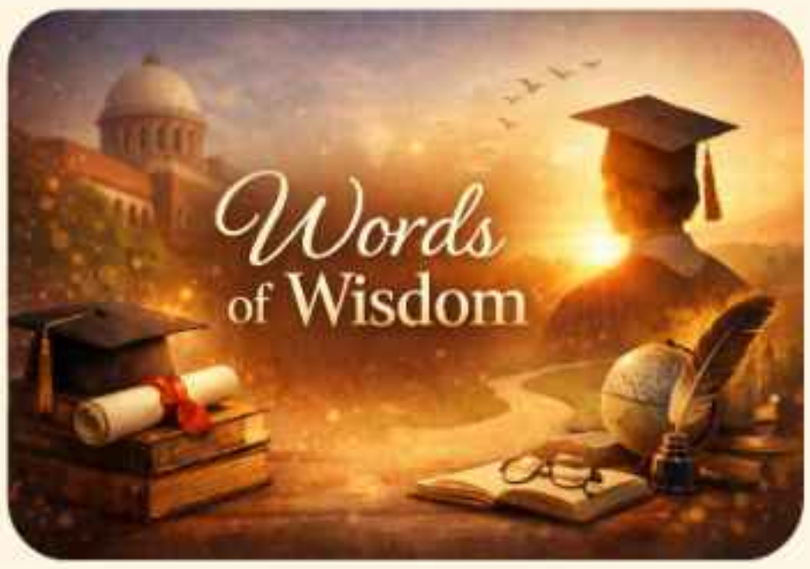
Once Students , Always Mentors





We are delighted to share an inspiring message from **Dr. Charanjeet Singh**, a proud alumnus and PhD graduate of our department, who offers valuable guidance and encouragement to students embarking on their academic journey at Panjab University.

" Dear Students,
Congratulations on embarking on this exciting phase of your life at Panjab University. University is a time for learning, growth, and self-discovery. Although you may encounter challenging moments, remember that they are simply part of the journey and will aid you in uncovering your true potential.

Take every opportunity to learn, meet new people, and explore what excites you. The friendships you cultivate here will support you, and the lessons you learn will remain with you long after graduation. Don't be afraid to try new things, make mistakes, and grow from them. Success isn't solely about getting things right; it's about embracing the journey.

Enjoy the journey, create lasting memories, and, most importantly, stay true to yourself. The world is eager for your unique talents, so go out there and make something extraordinary!

Wishing you all the best on this exciting journey ahead! "
Dr. Charanjeet Singh

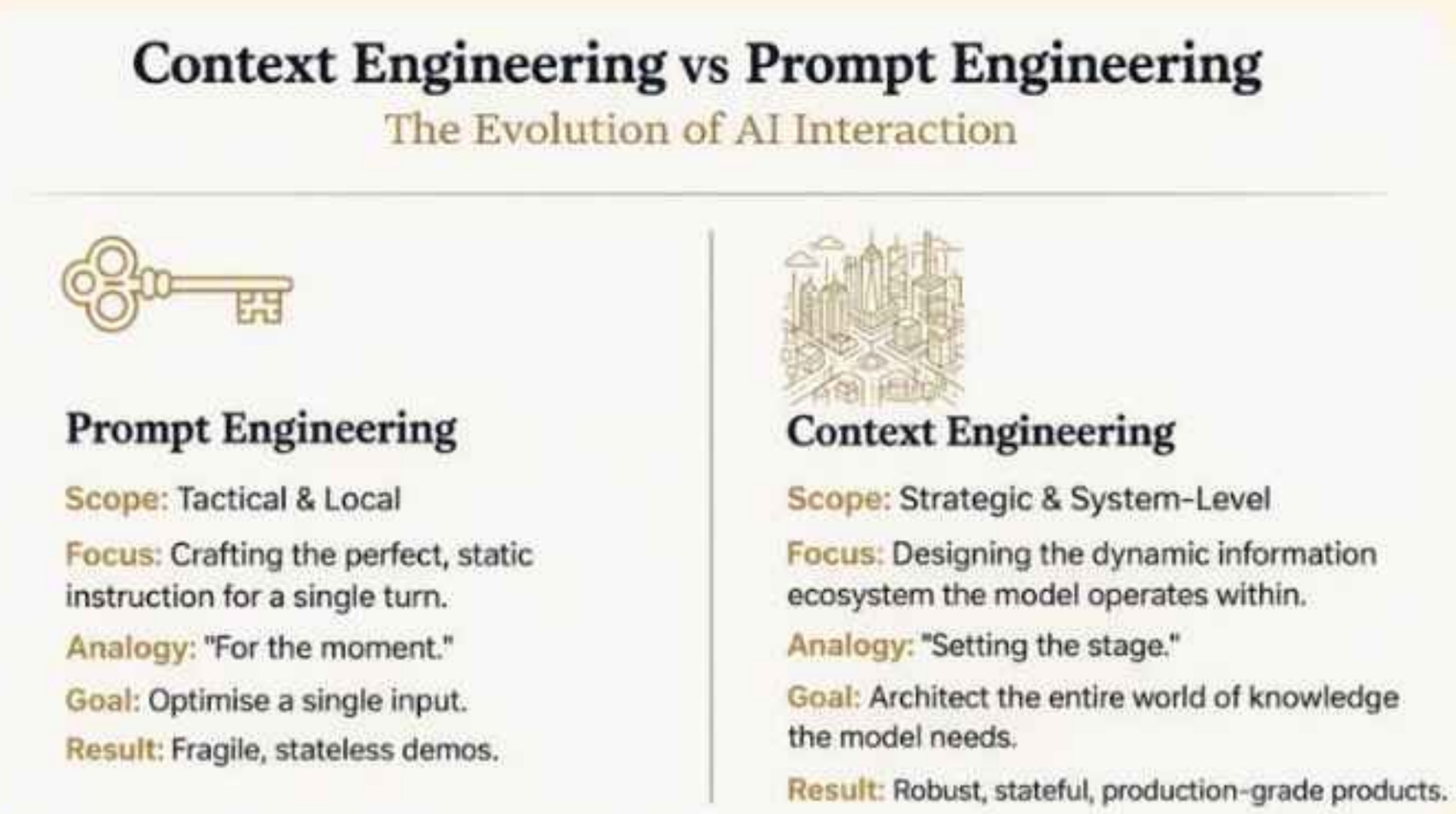
-Anshit Sehgal MCA-I(M) & Arnav Khantwal MCA-I(M)

Context Engineering vs Prompt Engineering : The future of AI Interaction

In the beginning of generative AI, the main way to talk to large language models (LLMs) was through prompt engineering, which involved writing clear, specific instructions to get a certain response. But as AI systems have evolved from basic tools to intricate, multi-step agents, the constraints of tactical prompting have been supplanted by a more advanced, systems-level discipline termed context engineering.

Prompt engineering is basically the skill of asking the right question. It focuses on how to write text in a way that will help the model do a specific task right away, including the words, structure, and formatting. Most people think of it as a stateless, single-turn practice that doesn't work well with dynamic, industrial-strength applications.

Context engineering, on the other hand, is the science of making sure the AI has the right information to answer those questions in a good way. It is officially defined as the systematic discipline of designing, orchestrating, and optimizing the complete information ecosystem provided to an LLM at inference time. This change means going from "what you say" to a model that takes into account "everything else the model sees," such as memory, retrieved documents, and tools that are available.



The difference between these two fields is not just a matter of language; it shows a big change in the structure of things.

- Scope : Prompt engineering works with just one input-output pair, while context engineering works with a "context stack" that has many layers.
- Mindset : People often compare prompting to creative writing or "prompt alchemy," and they see context engineering as systems design or software architecture.
- Tools : Prompt engineering uses text editors and testing over and over again, while context engineering uses Retrieval-Augmented Generation (RAG), vector databases, memory modules, and API chaining frameworks.
- Aim: The main goal of context engineering is to make sure that performance is consistent, reliable, and scalable across different users and sessions, which makes the prompt less of a burden.

A constructed context is a combination of several important information streams that help the model stay grounded in the real world:

- System instructions : Setting the AI's personality, rules, and limits.
- Memory : This includes both short-term (like the history of a conversation) and long-term (like user preferences or facts stored in vector databases).
- Retrieved Knowledge (RAG) : Adding new, outside, or proprietary data that wasn't part of the model's original training.
- Tool Definitions and Outputs : Information about the APIs that are available and what data they return when they are run.

For businesses, the value of AI is becoming more and more clear, not by changing the model, but by fixing the context. Even the best models can fail or "hallucinate" if they don't have a full picture of the world. Context engineering gives you the framework you need to make sure that the information is correct and tailored to you. Also, studies show that context engineering can lead to big savings, with costs dropping by 10% to 30% compared to prompting strategies that don't work well. The "Context-as-a-Compiler" Analogy: To understand this paradigm shift, one might use the analogy of a computer compiler. In this view, the LLM is a new kind of compiler that translates human intent into executable actions. The prompt is the source code, but the context represents the libraries, dependencies, and environment variables required for the compiler to produce a correct, non-hallucinated binary. Just as a traditional compiler fails if a function is not declared, an LLM will struggle if the necessary "declarations" are missing from its context window.

– Ashish Yadav (Research Scholar)

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